Assignment-03

Name : Daniel Jesuraji. J.

Reg No : 192565085.

Course code : CSA5117.

Course Name: Cryptography and Network Security

98/100

Q). Explain the principle of the digital signature standard [DSS]. Which algorithm does it rely on?

The digital signature standard (DSS) is a developed by the National Institute of Standard and Technology (NIST) for creating and verify the digital signatures. A digital signature provides three important security services.

The basic principle of DSS is that sender uses private key to generate a signature for a message, while the receiver uses the corresponding public key to verify the signature.

Working of DSS:-

⇒ The sender creates a message that need to be transmitted securely.

⇒ A hash function is applied to the message to produce a fixed-size message digit.

⇒ The sender uses the private key to generate a digital signature from the message digest.

⇒ The message and digital signature are sent to the receiver.

⇒ The receiver uses the sender's public key to verify the signature.

⇒ If the verification is successful, the receiver can confirm that the message from the claimed sender.

Sender
↓
Message
↓
Hash function.
↓
Message digest
↓
Private key.
↓
Digital signature.
↓
Message + signature.
↓
Receiver.
↓
Public key verification.
↓
Valid / invalid.

b) In ElGmail digital signature, Explain how the random value k affects the signature security.

In the ElGmail digital signature scheme a random value k is selected during the signature generature process. This value plays an important role in protecting the private key and ensuring that each signature is secure.

The value k must be:

* Random.
* Unpredictable.
* kept secret
* Different for every signature.

Importance of k:

Secure Random k.
↓
Unique signature.
↓
Private key protected.
↓
Secure ElGamal signature.

weak / Reused k.
↓
Signature information leaked.
↓
Private key may Be Recovered.
↓
Signature system compromised.

c). The schnorr protocol relies on the hardness of the discrete logarithm problem, Explain how the challenge - respone mechanism works

The schnorr protocol is a challenge - response authentication protocol based on the difficulty solving the discrete logarithm problem

It allows a prover to demonstrate knowledg of a secrete without directly revealing that secret to the verifier.

Commitment:

The prover selects a random and uses it to calculate a commitment.

Prover.
↓
Commitment.
↓
Verifier.

Challenge:

After receiving the commitment the verifier generates a random challenge and send it back to the prover.

Power.
↑
challenge
Verifier.

Response:

The prover generate a response using the secret value, the random value used in the commitment and the challenge received from the verifier.

commitment + challenge + Response.

↓

Verification

↙ ↘

valid invalid

↓ ↓

Accepted Rejected.

d). Compare Elgamal and schnorr digital signature schemes.

feature	Elgamal Digital signature	schnorr digital signature.
Basic principle.	Uses public-key digital signature.	Uses challenge-response based signing.
Security basis.	Discrete logarithm Problem.	Discrete logarithm Problem.
signature generation.	Relatively more computational operations.	Generally more efficient.
signature size	Relatively large	Generally smaller.
Efficiency	Moderate	Generally high.
Main concern	secure generation and non reuse of k.	secure implementation of challenge and response.
Suitable for	General public-key signatures.	Efficient authentication and signatures.

Conclusion:-

Both ElGamal and schnorr digital signature provides authentication and integrity based on the discrete logarithm Problem. ElGamal is a well establised signature scheme but requires careful management of the random value k. schnorr provide a simpler and generally more efficient approach through its challenge-response mechanism. Therefore, schnorr is generally preferred when efficiency, compact signatures, and efficient verification are important. while ElGamal remains an important discrete logarithm based digital signature scheme.

* * *